Online Polycube Packing with Deep Reinforcement Learning

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INTRODUCTION & MOTIVATION

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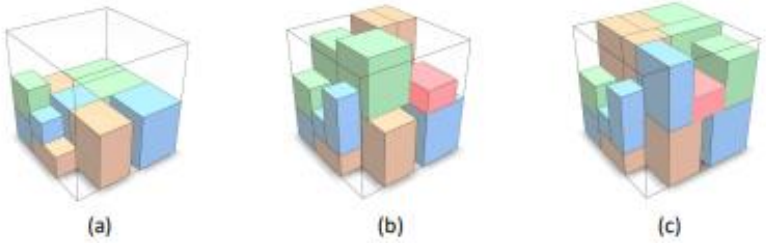
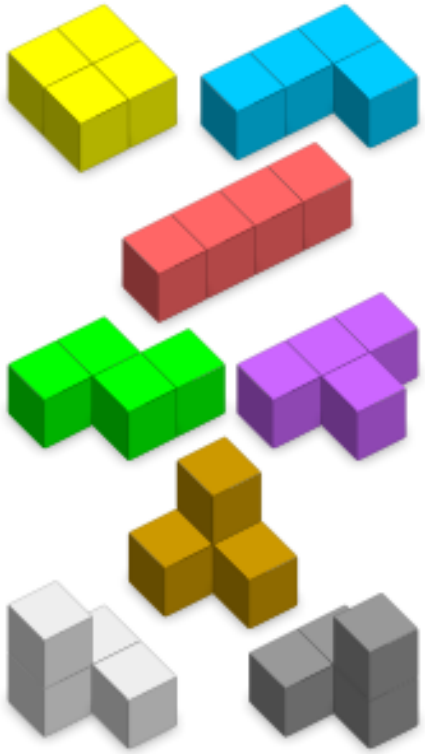
Introduction – Polycube Packing

- Packing Problems



Regular Shapes



Polycubes

Irregular Shapes

Introduction – Online Packing

- Get Polycubes Sequentially
- Heuristics vs. Learning
- Restricting the Action Space

Research Questions

1. Can online polycube packing be solved using deep reinforcement learning?

2. How does the performance of deep reinforcement learning compare to traditional heuristics?

3. Can the performance of deep reinforcement learning be improved by using heuristics to restrict the action space?

Methods – Markov Decision Process

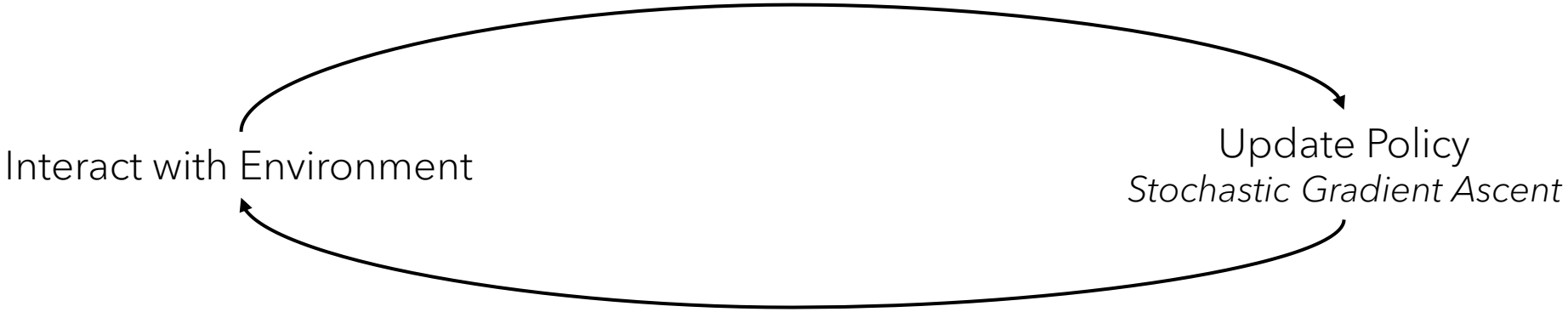
- States: *container + polycube*
- Actions: *position + orientation*
- Transition Probability Function: *stochastic*
- Reward Function: *number of polycubes packed*

Methods – Deep Reinforcement Learning

- Large State and Action Space
- Stochastic Environment

On-Policy, Model-Free

- Proximal Policy Optimization (PPO)



Experiments

- PPO vs. Greedy with Heuristics

PPO-D	PPO-O	PPO-H
Default Hyperparameters	Optimized Hyperparameters	Restricted Action Space

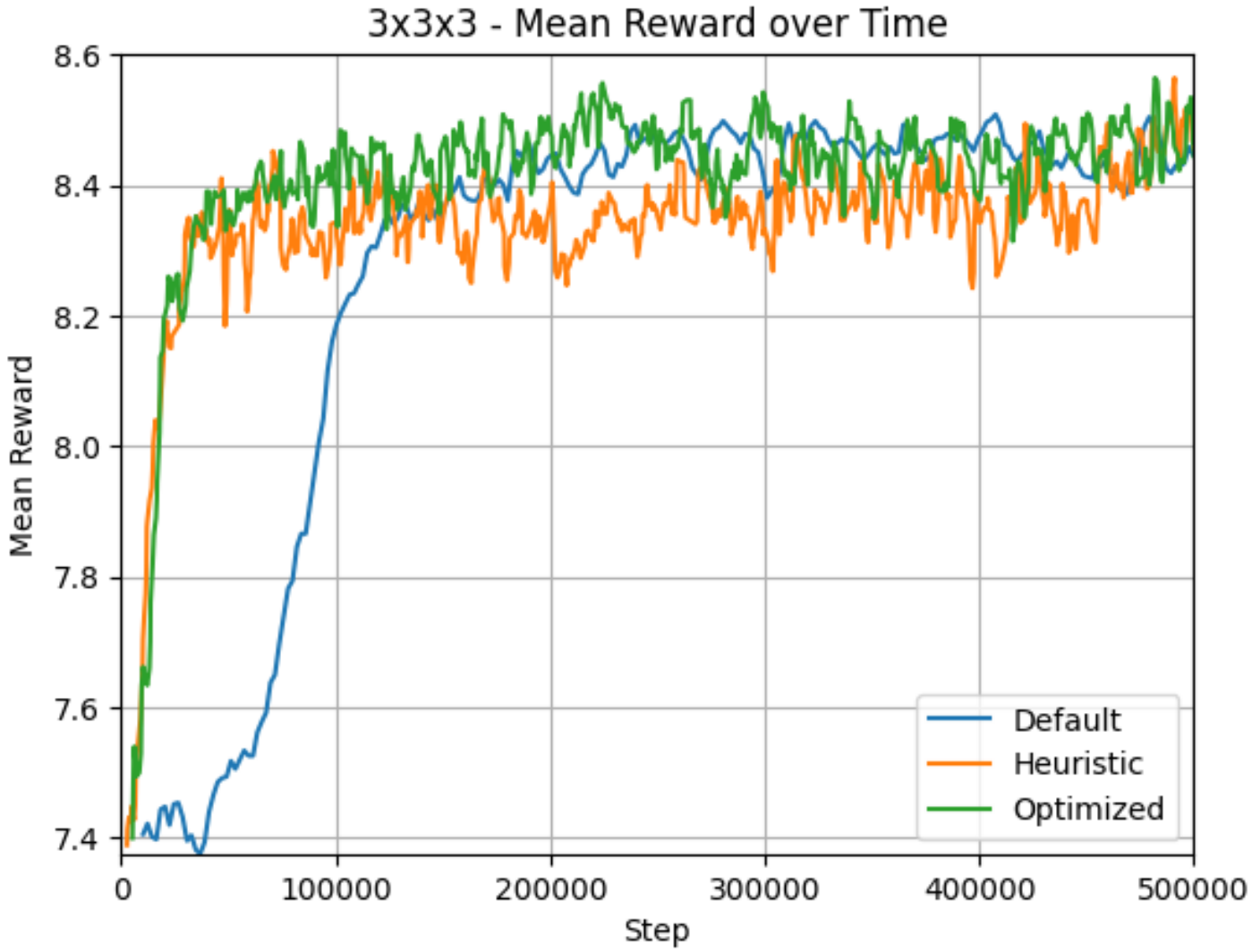
- Container Sizes: *3x3x3, 5x3x3, 4x4x4, 6x4x4*

Experiment 1

- Training PPO Agents
  - Fixed number of steps
  - Until convergence

- Mean Reward

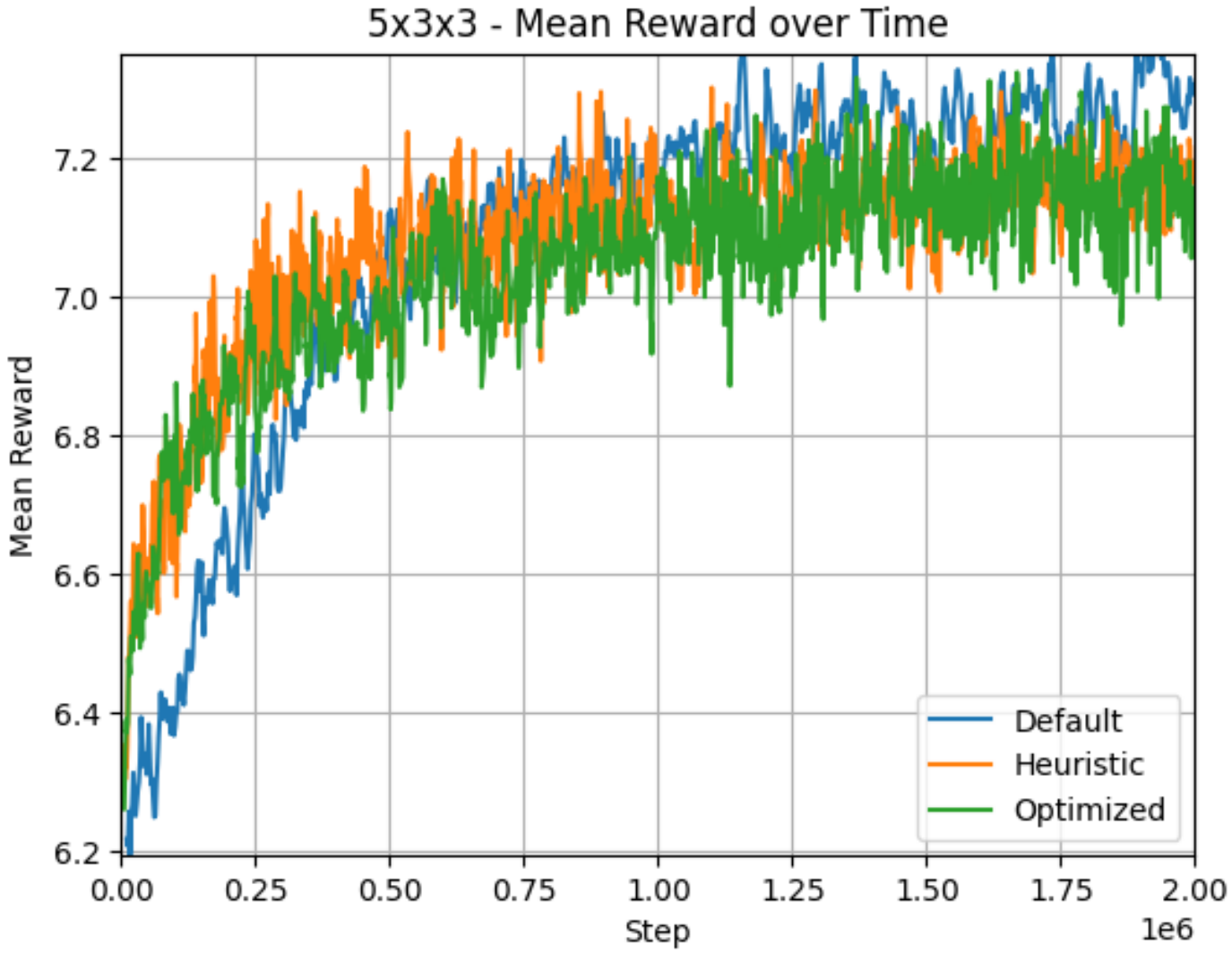
- Entropy Loss



PPO-D
Default Hyperparameters

PPO-O
Optimized Hyperparameters

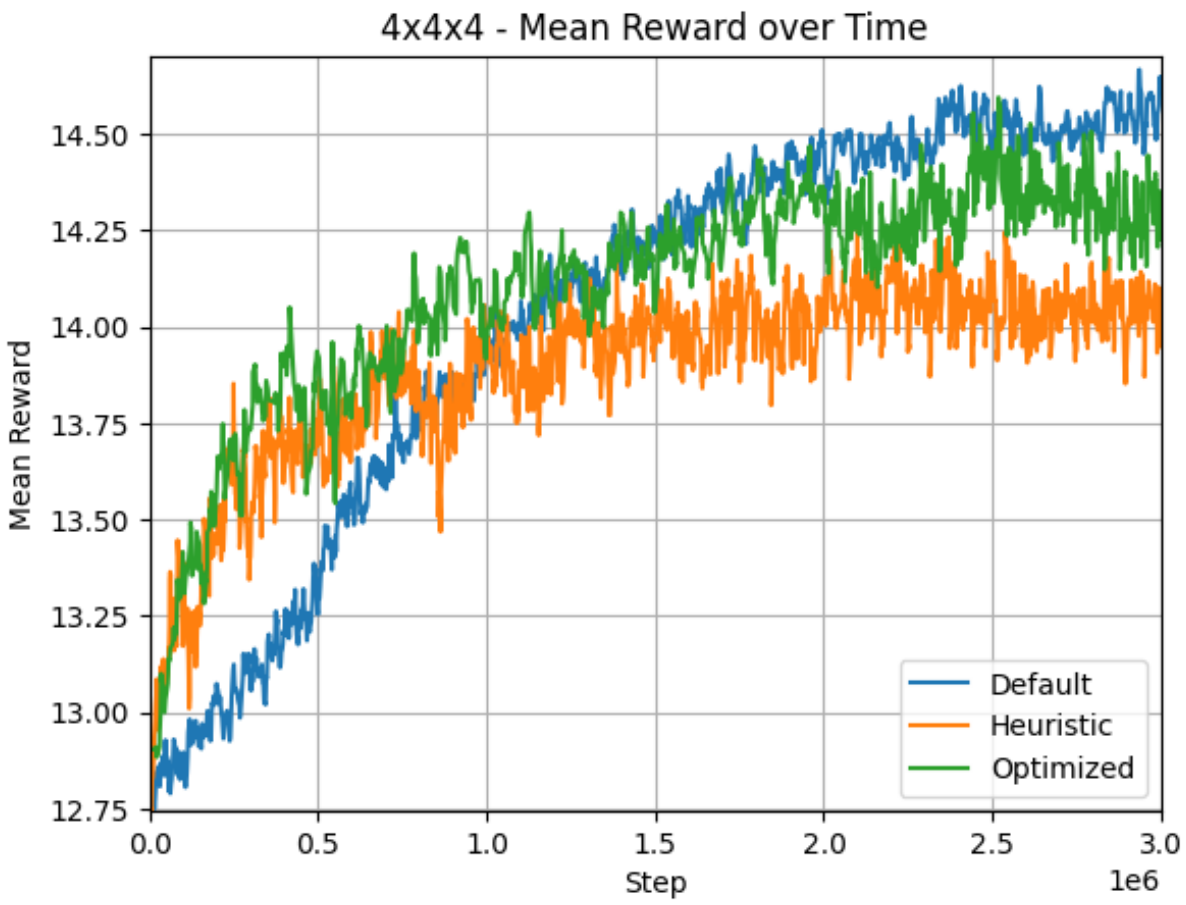
PPO-H
Restricted Action Space

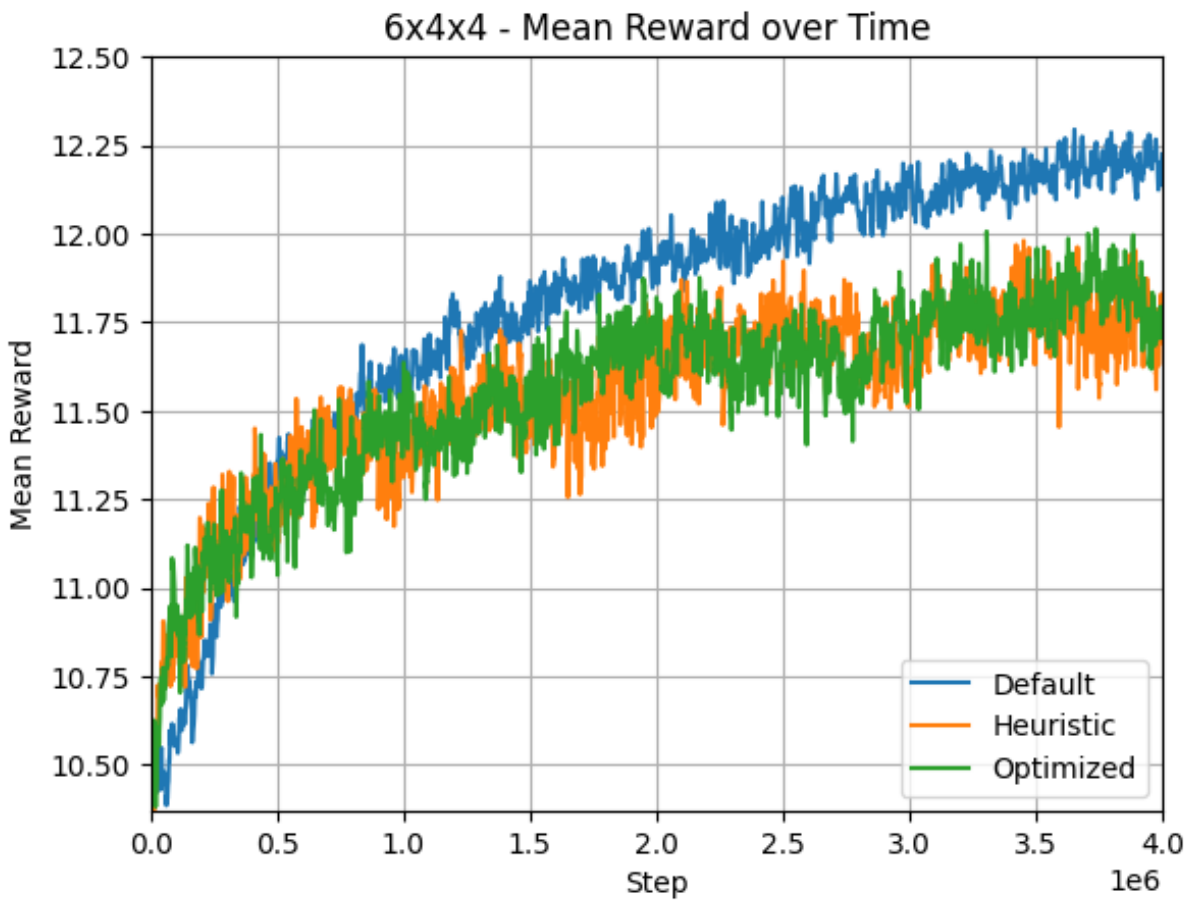


PPO-D
Default Hyperparameters

PPO-O
Optimized Hyperparameters

PPO-H
Restricted Action Space



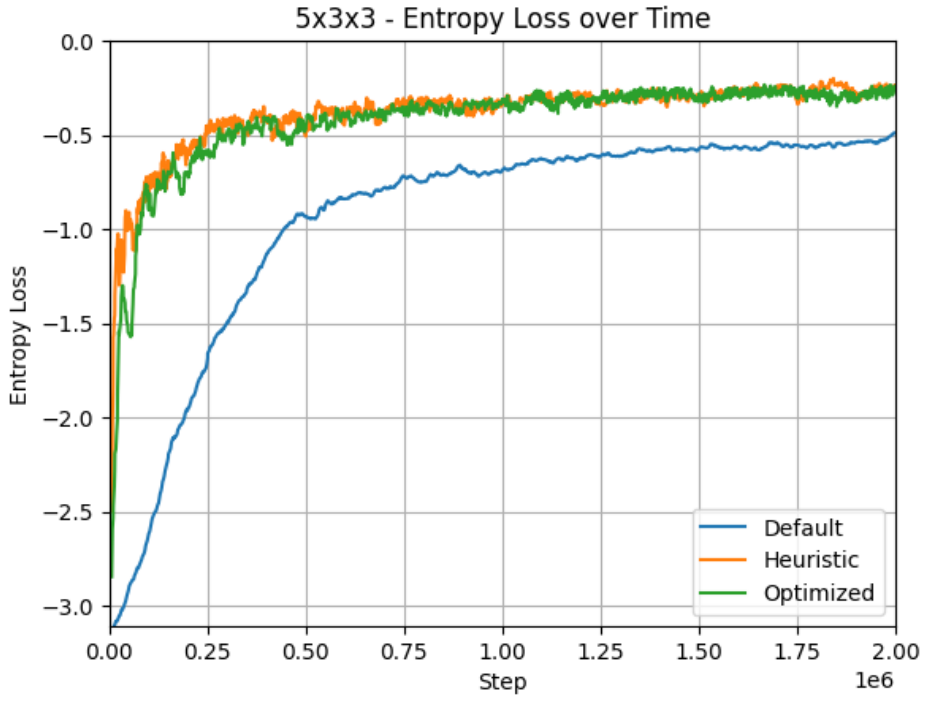


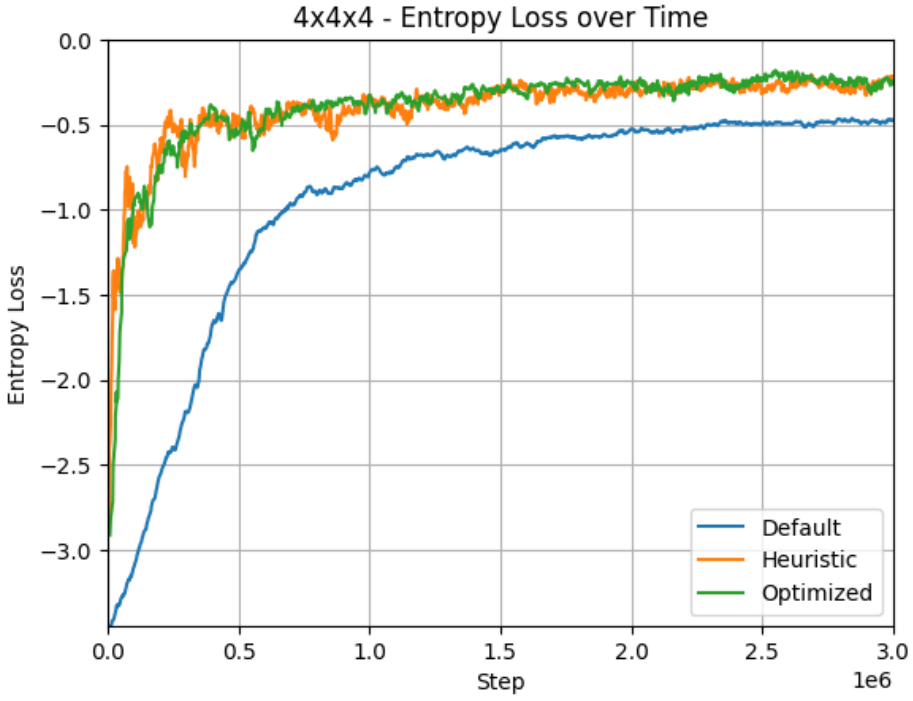
PPO-D
Default Hyperparameters

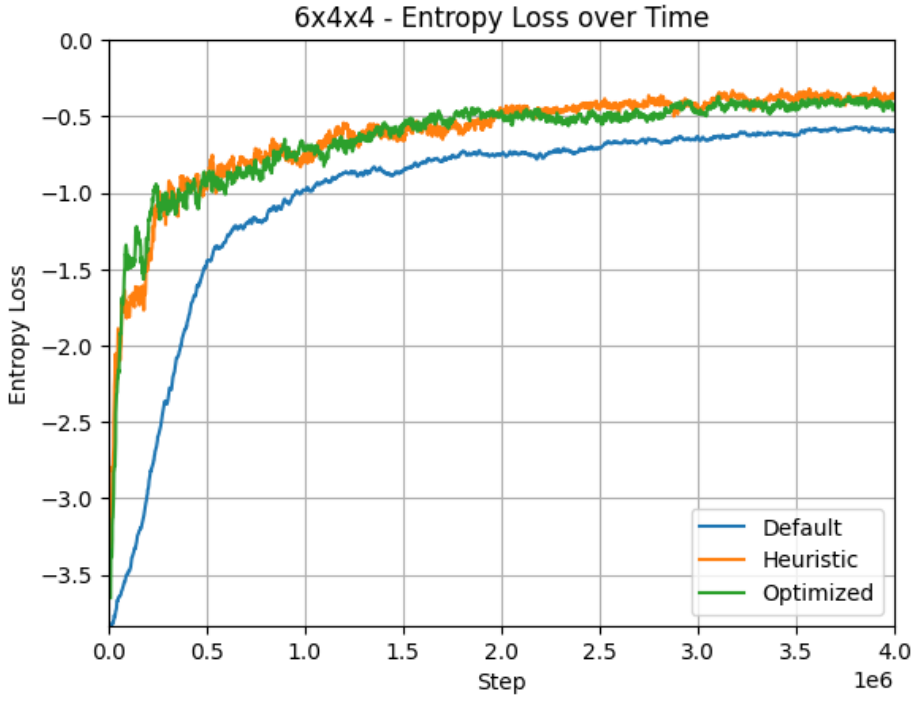
PPO-O
Optimized Hyperparameters

PPO-H
Restricted Action Space

3x3x3 - Entropy Loss over Time
Entropy Loss
Step
Default
Heuristic
Optimized


5x3x3 - Entropy Loss over Time
Entropy Loss
Step
1e6
Default
Heuristic
Optimized


4x4x4 - Entropy Loss over Time
Entropy Loss
Step
1e6
Default
Heuristic
Optimized

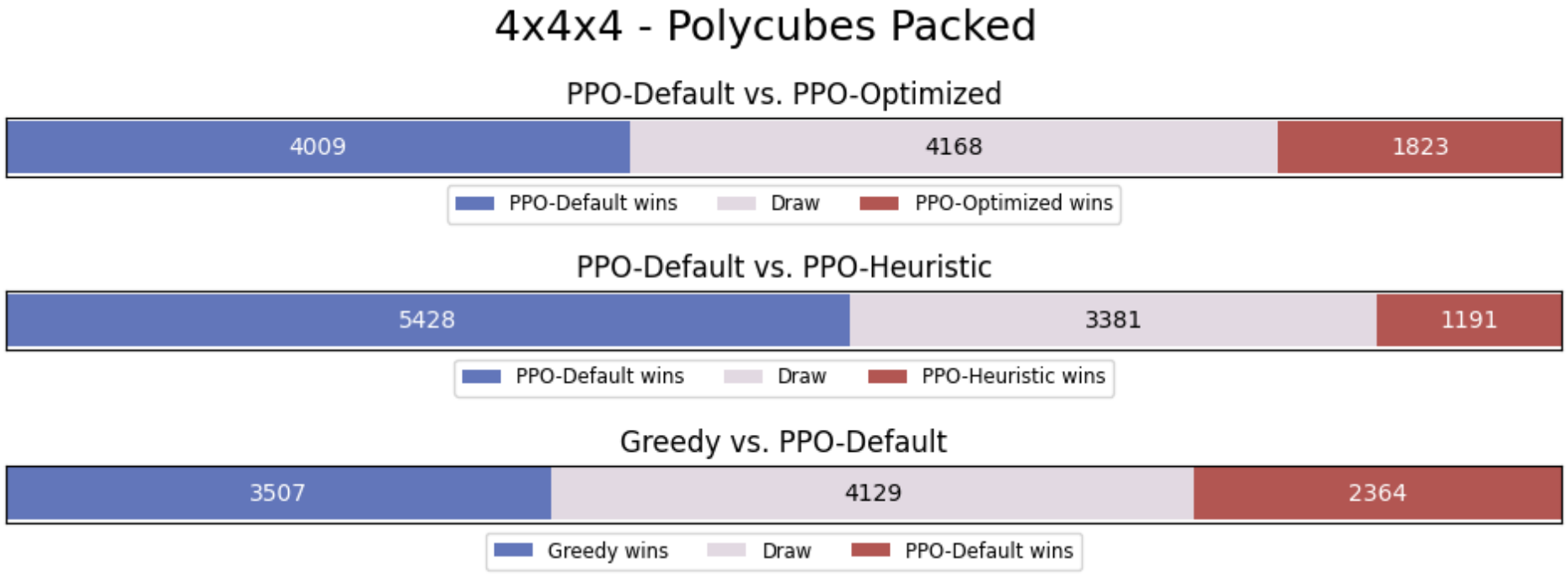

6x4x4 - Entropy Loss over Time
Entropy Loss
Step
1e6
Default
Heuristic
Optimized

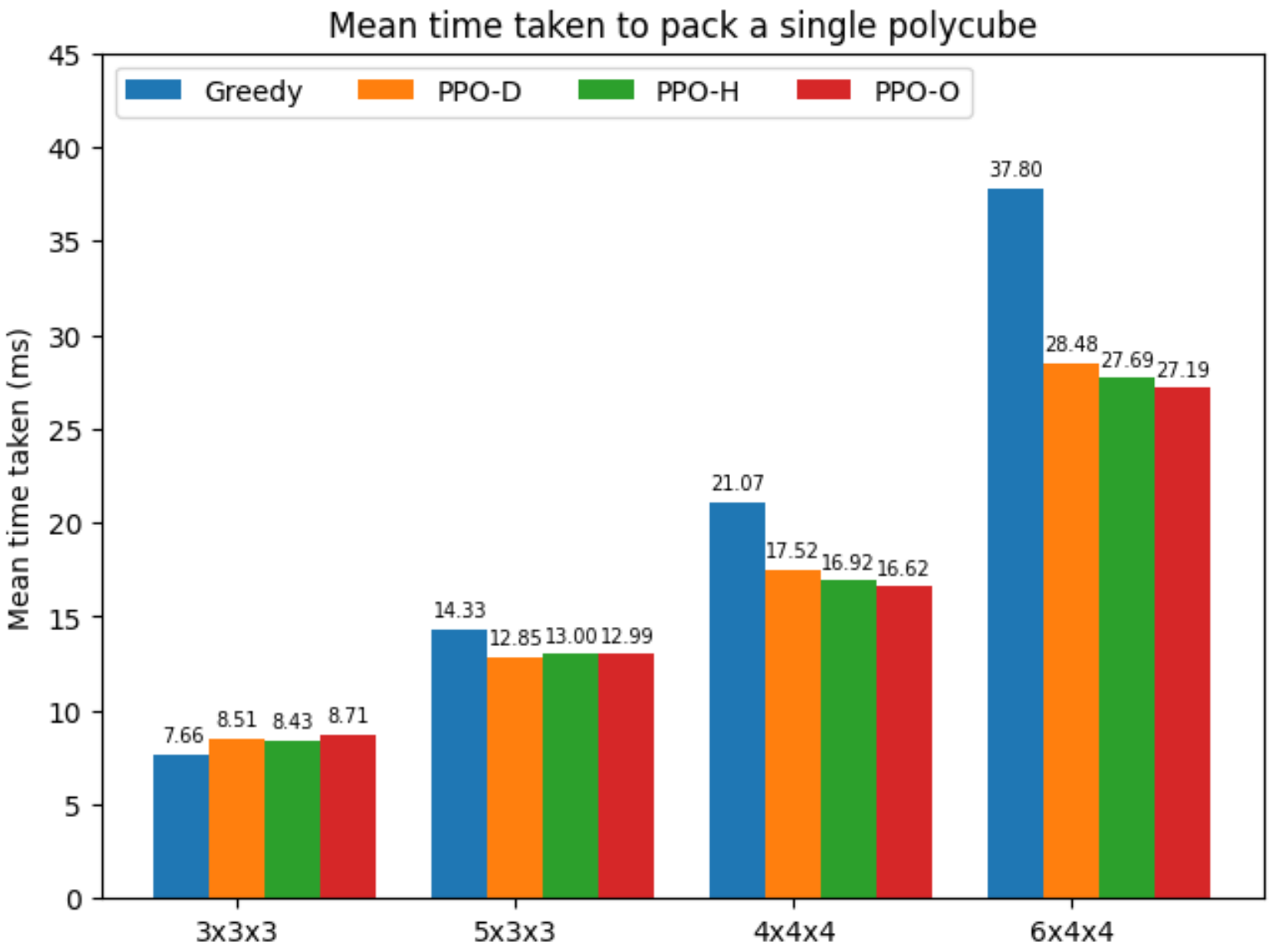
Experiment 2

- Measuring Performance
  - Head-to-Head
  - 10,000 Sequences

- Number of Polycubes Packed

- Time Taken


4x4x4 - Polycubes Packed
PPO-Default vs. PPO-Optimized
4009
4168
1823
PPO-Default wins
Draw
PPO-Optimized wins
PPO-Default vs. PPO-Heuristic
5428
3381
1191
PPO-Default wins
Draw
PPO-Heuristic wins
Greedy vs. PPO-Default
3507
4129
2364
Greedy wins
Draw
PPO-Default wins


Mean time taken to pack a single polycube
Greedy
PPO-D
PPO-H
PPO-O
Mean time taken (ms)
0
5
10
15
20
25
30
35
40
45
7.66
8.51
8.43
8.71
14.33
12.85
13.00
12.99
21.07
17.52
16.92
16.62
37.80
28.48
27.69
27.19
3x3x3
5x3x3
4x4x4
6x4x4

Conclusion

1. Can online polycube packing be solved using deep reinforcement learning? - Yes

2. How does the performance of deep reinforcement learning compare to traditional heuristics? - Faster, but not as good

3. Can the performance of deep reinforcement learning be improved by using heuristics to restrict the action space? - No

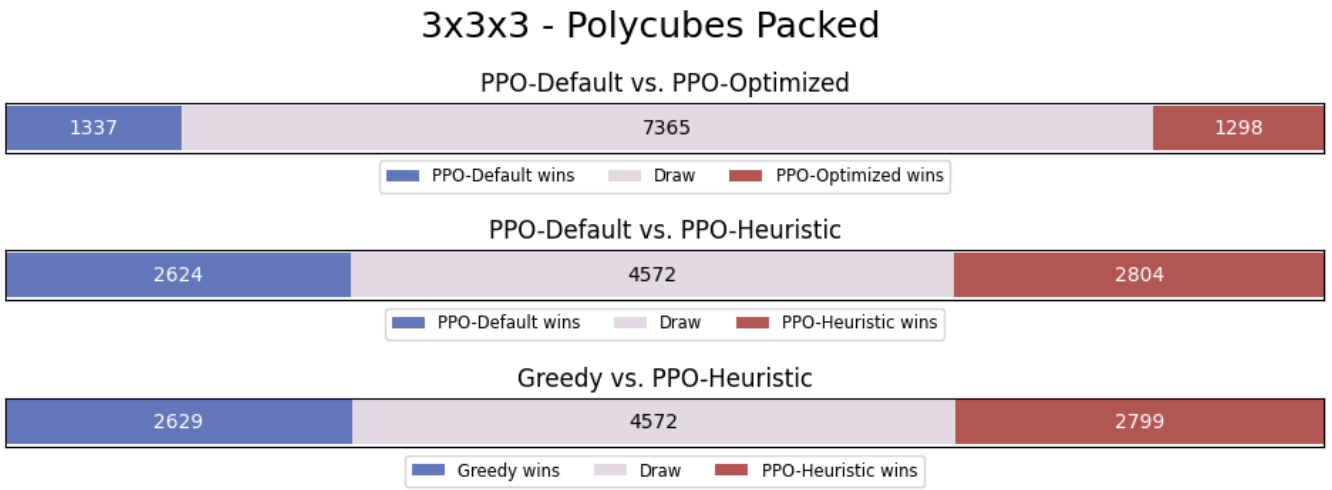
Limitations

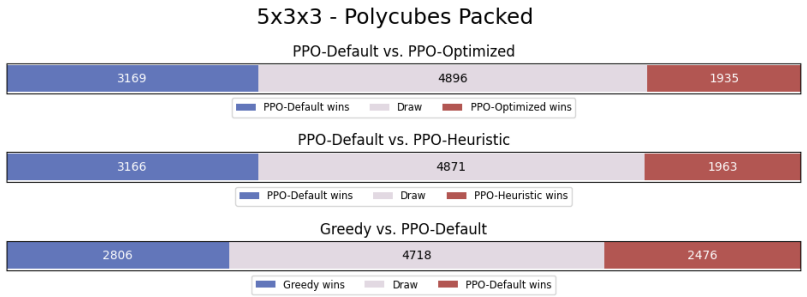
- Computational Cost
- Larger Container Sizes
- Hyperparameter Tuning

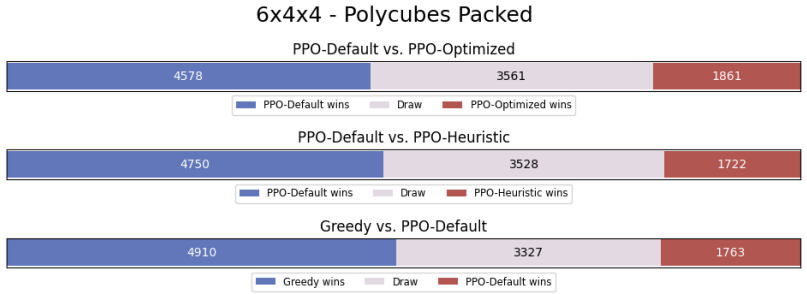
References:

- P. Louridas, Real-World Algorithms (Title Slide)
- H. Zhao et al., Online 3D Bin Packing with Constrained Deep Reinforcement Learning (Introduction – Polycube Packing)
- H. Zhao et al., Learning Physically Realizable Skills for Online Packing of General 3D Shapes (Introduction – Polycube Packing)
- Cmglee, Wikipedia: Polycube (Introduction – Polycube Packing)

Appendix


3x3x3 - Polycubes Packed
PPO-Default vs. PPO-Optimized
1337
7365
1298
PPO-Default wins
Draw
PPO-Optimized wins
PPO-Default vs. PPO-Heuristic
2624
4572
2804
PPO-Default wins
Draw
PPO-Heuristic wins
Greedy vs. PPO-Heuristic
2629
4572
2799
Greedy wins
Draw
PPO-Heuristic wins


5x3x3 - Polycubes Packed
PPO-Default vs. PPO-Optimized
3169
4896
1935
PPO-Default wins
Draw
PPO-Optimized wins
PPO-Default vs. PPO-Heuristic
3166
4871
1963
PPO-Default wins
Draw
PPO-Heuristic wins
Greedy vs. PPO-Default
2806
4718
2476
Greedy wins
Draw
PPO-Default wins


6x4x4 - Polycubes Packed
PPO-Default vs. PPO-Optimized
4578
3561
1861
PPO-Default wins
Draw
PPO-Optimized wins
PPO-Default vs. PPO-Heuristic
4750
3528
1722
PPO-Default wins
Draw
PPO-Heuristic wins
Greedy vs. PPO-Default
4910
3327
1763
Greedy wins
Draw
PPO-Default wins

PPO Training Times

	PPO-D	PPO-O	PPO-H
3x3x3	1.5 hr	1.5 hr	2 hr
5x3x3	8.5 hr	6.5 hr	12 hr
4x4x4	14 hr	14 hr	20 hr
6x4x4	32 hr	32 hr	45 hr

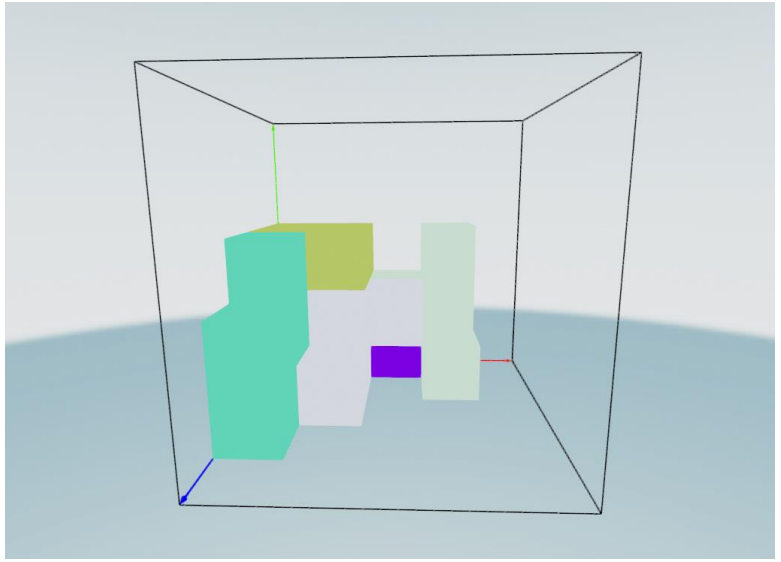
parameter	value
learning rate	0.0003
number of steps	2048
batch size	64
update epochs	10
gamma	0.99
GAE lambda	0.95
clip range	0.2
entropy coefficient	0
value function coefficient	0.5
max gradient norm	0.5

Default Hyperparameters

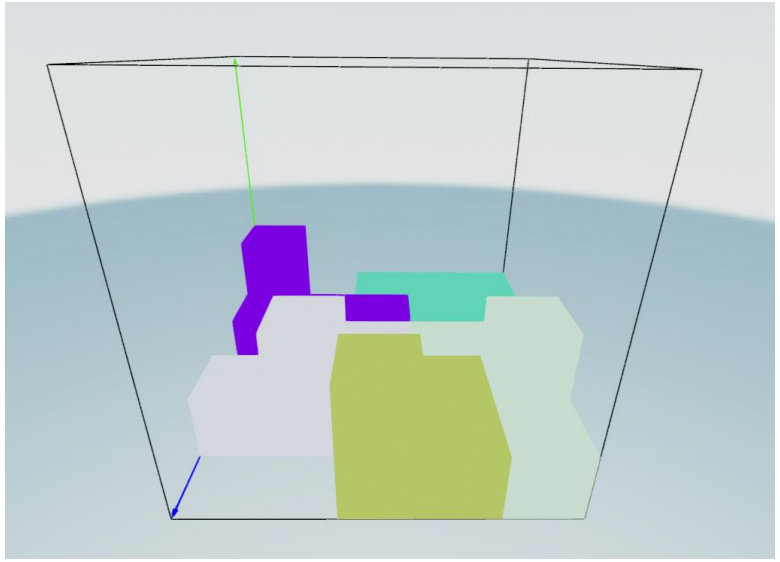
parameter	value
learning rate	0.0005
number of steps	64/128/256*
entropy coefficient	0.005
heuristics	BLBF + HAPE
n	50/75/100/200*

Optimized Hyperparameters

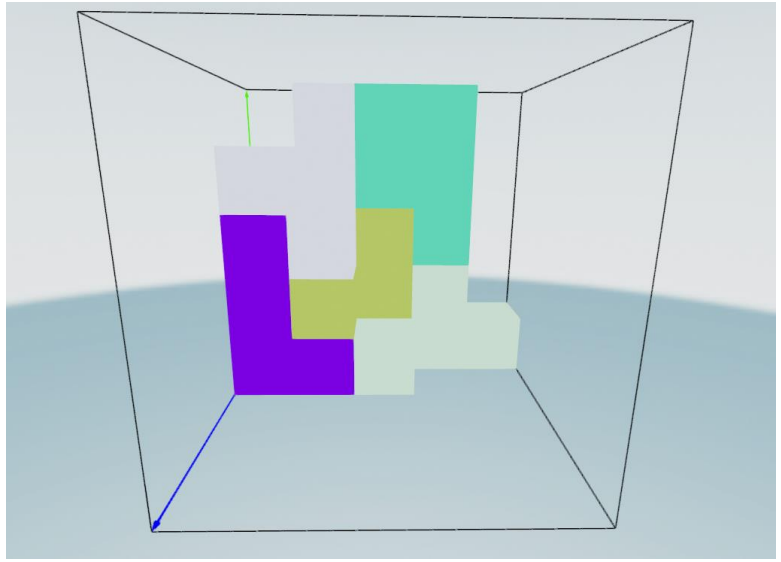
Heuristics



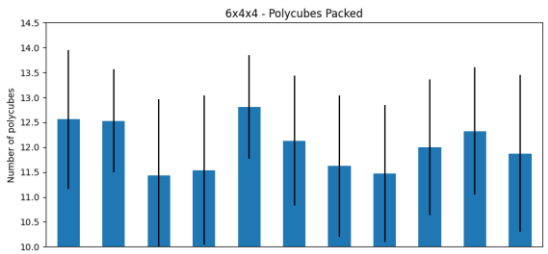
BLBF

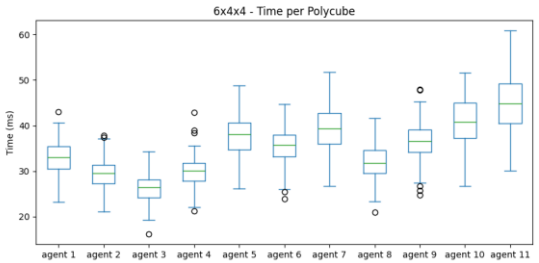


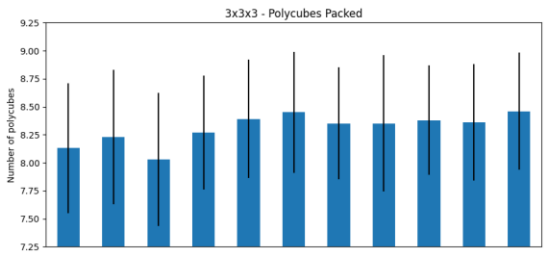
HAPE

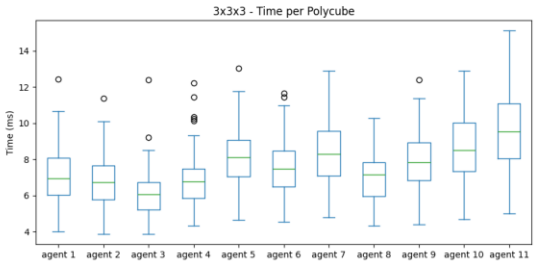


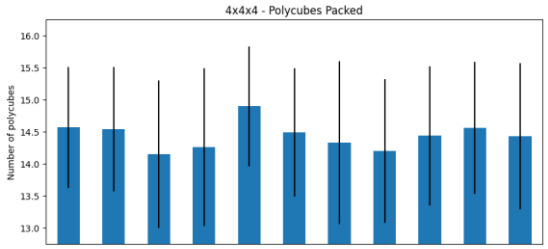
Heightmap Minimization

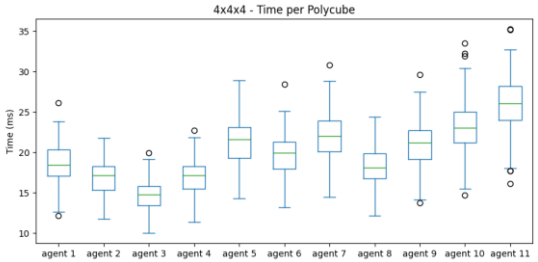

6x4x4 - Polycubes Packed
Number of polycubes
14.5
14.0
13.5
13.0
12.5
12.0
11.5
11.0
10.5
10.0

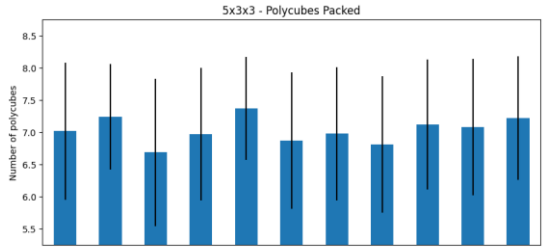

6x4x4 - Time per Polycube
Time (ms)
60
50
40
30
20
agent 1
agent 2
agent 3
agent 4
agent 5
agent 6
agent 7
agent 8
agent 9
agent 10
agent 11

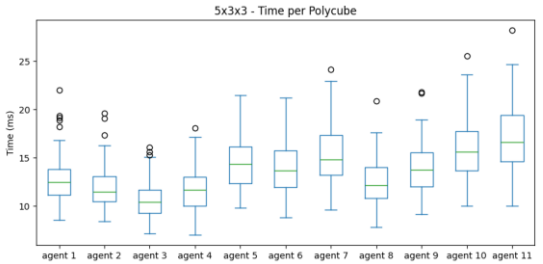

3x3x3 - Polycubes Packed
Number of polycubes
9.25
9.00
8.75
8.50
8.25
8.00
7.75
7.50
7.25


3x3x3 - Time per Polycube
Time (ms)
14
12
10
8
6
4
agent 1
agent 2
agent 3
agent 4
agent 5
agent 6
agent 7
agent 8
agent 9
agent 10
agent 11


4x4x4 - Polycubes Packed
Number of polycubes
16.0
15.5
15.0
14.5
14.0
13.5
13.0


4x4x4 - Time per Polycube
Time (ms)
35
30
25
20
15
10
agent 1
agent 2
agent 3
agent 4
agent 5
agent 6
agent 7
agent 8
agent 9
agent 10
agent 11


5x3x3 - Polycubes Packed
Number of polycubes
8.5
8.0
7.5
7.0
6.5
6.0
5.5


5x3x3 - Time per Polycube
Time (ms)
25
20
15
10
agent 1
agent 2
agent 3
agent 4
agent 5
agent 6
agent 7
agent 8
agent 9
agent 10
agent 11